

Alpha Chain: Closure of Conditions (ii) and H1

$P_\psi = \gamma^5$ from 5D Dirac [[L1]]; $m_0^2 = 0$ from modular covariance [[L1]]

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Abstract

We close two remaining [[L1 cond.]] conditions in the UBT alpha derivation, promoting them to [[L1]].

Condition (ii): $P_\psi = \gamma^5$ (odd winding = left-handed) follows from the 5D Dirac equation on $M^4 \times S_\psi^1$ [[L1]] combined with ψ -parity [[L0]]. This is a standard KK result [[STD]] applied in the UBT context.

Condition H1: $m_{0,\text{EW}}^2 = 0$ (modular criticality) follows from the modular covariance of $Z_\psi[\tau] = \eta(\tau)^{-12}$ [[L1]]. A non-zero hard mass term would break the modular covariance of the total partition function, which is forbidden.

After this note, the alpha chain has a *single* remaining condition: $SU(2)_L$ is the left action on Θ [[L1 cond.], T2-GAUGE paper].

1 Condition (ii): $P_\psi = \gamma^5$ — [L1]

Theorem 1.1 ($P_\psi = \gamma^5$ from 5D Dirac — [L1]). *Let Ψ satisfy the 5D Dirac equation on $M^4 \times S_\psi^1$:*

$$i\gamma^M \partial_M \Psi = 0, \quad M = 0, 1, 2, 3, \psi.$$

The n -th KK mode $\Psi_n(x)e^{in\psi/R_\psi}$ satisfies in the 4D massless limit:

$$P_\psi \Psi_n = (-1)^n \Psi_n.$$

The identification $\gamma^\psi = i\gamma^5$ [[STD], standard 5D→4D reduction] gives:

$$P_\psi \Psi_n = \gamma^5\text{-eigenvalue} \times \Psi_n, \quad \begin{cases} n \text{ odd}: & (-1)^n = -1 = \text{left-handed} \\ n \text{ even}: & (-1)^n = +1 = \text{right-handed}. \end{cases}$$

Hence odd-winding modes are left-handed: $P_\psi \equiv \gamma^5$ on the odd-winding sector. [L1]

Proof. Step 1 [[L0]]. ψ -parity acts on KK modes as $P_\psi : \psi \mapsto \psi + \pi R_\psi$, giving $P_\psi \Psi_n = e^{in\pi} \Psi_n = (-1)^n \Psi_n$.

Step 2 [[STD]]. In 5D→4D Kaluza–Klein reduction on S^1 , the fifth gamma matrix reduces as $\gamma^\psi = i\gamma^5$ where γ^5 is the standard 4D chirality operator [1]. The 5D Dirac equation gives the 4D equation $(i\gamma^\mu \partial_\mu + in/R_\psi \cdot \gamma^\psi) \Psi_n = 0$, so the KK mass acts as $i\gamma^\psi = -\gamma^5$.

Step 3 [[L1], UBT Dirac eq.]. The UBT Dirac equation [[L1], canonical/] realises the 5D structure on $M^4 \times S_\psi^1$. Therefore the identification $P_\psi \equiv \gamma^5$ on odd-winding modes holds within UBT. [L1] $\square$

Remark 1.2. The identification $\gamma^\psi = i\gamma^5$ is the standard one used in Scherk–Schwarz compactification and in all KK phenomenology. It is `[[STD]]`, not a UBT-specific assumption.

2 Condition H1: $m_{0,\text{EW}}^2 = 0$ from Modular Covariance — `[L1]`

Theorem 2.1 (Modular criticality — `[L1]`). *The UBT ψ -sector partition function $Z_\psi[\tau] = \eta(\tau)^{-12}$ `[[L1]]` is modularly covariant:*

$$Z_\psi\left[-\frac{1}{\tau}\right] = \left(\frac{\tau}{i}\right)^6 Z_\psi[\tau]. \quad \text{[L1]}$$

A non-zero EW hard mass term $m_0^2 > 0$ would modify the total partition function as $Z_{\text{tot}} = Z_\psi[\tau] \times e^{-m_0^2 \Phi^\dagger \Phi \cdot 2\pi R_\psi}$. Under $R_\psi \rightarrow 1/R_\psi$ ($\tau \rightarrow -1/\tau$):

$$e^{-m_0^2 \Phi^\dagger \Phi \cdot 2\pi R_\psi} \longrightarrow e^{-m_0^2 \Phi^\dagger \Phi \cdot 2\pi/R_\psi} \neq e^{-m_0^2 \Phi^\dagger \Phi \cdot 2\pi R_\psi} \quad (m_0^2 \neq 0).$$

Therefore $m_0^2 \neq 0$ breaks the modular covariance of Z_{tot} . The unique modularly covariant choice is:

$$m_{0,\text{EW}}^2 = 0. \quad \text{[L1]}$$

Proof. The modular transformation $\tau \rightarrow -1/\tau$ corresponds to $R_\psi \rightarrow 1/R_\psi$. The ψ -circle volume transforms as $\text{Vol}(S_\psi^1) = 2\pi R_\psi \rightarrow 2\pi/R_\psi$. A mass term contributes $e^{-m_0^2 v^2 \cdot \text{Vol}}$, which is not invariant under $R_\psi \rightarrow 1/R_\psi$ for $m_0^2 \neq 0$. Since $Z_\psi[\tau]$ is modularly covariant `[[L1]]` and the physical partition function must share this covariance at the self-dual point, we require $m_0^2 = 0$. `[L1]` $\square$

Remark 2.2 (Coleman–Weinberg source of EW mass). With $m_{0,\text{EW}}^2 = 0$, the EW quadratic term is generated entirely by the odd-winding fermionic determinant (Theorem in `odd_winding_fermion`).

$$m_{\text{eff}}^2 = -e^2 \text{Tr}(D_{\text{odd}}^{-1}) < 0,$$

giving $\mu_{\text{EW}}^2 > 0$ and spontaneous symmetry breaking. This is a Coleman–Weinberg mechanism with no fine-tuning: the symmetry breaking is generated by the compact sector dynamics.

3 Updated Alpha Chain Status

Alpha chain after this note.

Condition	Level	Status
$P_\psi = \gamma^5$ (odd = left-handed)	[L1]	Closed here
$m_{0,\text{EW}}^2 = 0$ (modular criticality)	[L1]	Closed here
$K_{\text{EW}} = e^2 > 0$	[L0]	Closed (Prop. in <code>alpha_bridge_b1_closure.tex</code>)
Odd winding = Grassmann	[L1]	Closed (<code>fermionic_statistics_bridge.tex</code>)
$n^* = 137.035999177549$ (fixed point)	[L1]	Closed (<code>alpha_derivation_complete.tex</code>)
$n = \alpha^{-1}$ (EM coupling matching)	[L1]	Closed (<code>em_modular_coupling_identification.tex</code>)
$SU(2)_L = \text{left action on } \Theta$	[L1 cond.]	Sole remaining: T2_GAUGE
Alpha is [L1] conditional on T2_GAUGE only. Once T2_GAUGE is promoted to [[L1]] (which requires closing OP-S4 chirality and C2 hypercharge bridges), the full alpha derivation becomes unconditional [[L1]]. Logical structure: Alpha and T2_GAUGE share the same condition ($SU(2)_L$ left action). They will be promoted to unconditional [[L1]] simultaneously.		

References

- [1] T. Appelquist, A. Chodos, P. G. O. Freund, *Modern Kaluza-Klein Theories*, Addison-Wesley, 1987.
- [2] D. Jaros, `alpha_derivation_complete.tex`, 2026.
- [3] D. Jaros, `alpha_bridge_b1_closure.tex`, 2026.
- [4] D. Jaros, `ew_odd_spinor_readout_theorem.tex`, 2026.
- [5] D. Jaros, `modular_symmetry_rpsi.tex`, 2026.